

Microeconomic Theory: TA Session

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Homework 5, Q2

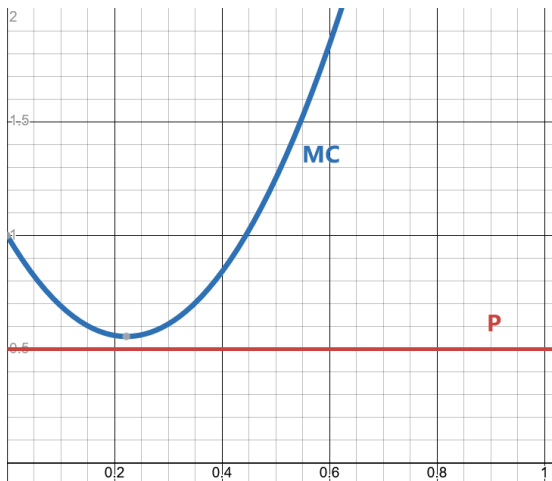
The total cost function for Professor D's Grape-nuts Hut is $C(q) = 3q^3 - 2q^2 + q + 2$, where q is the customers served per week. The market price of a Grape-nuts breakfast is a fixed p . Professor D's can serve as many customers per week as it chooses at that price.

- ▶ If $p = 0.5$, how many customers should it serve? What will be its maximized profit? Should it shut down immediately, or remain open for the time being? If the price and costs function are NOT going to change, should it shut down or remain open in the “long run”?

Answer: We attempt to solve the firm's profit maximization problem $P = MC \Rightarrow 9q^2 - 4q + 1 = 0.5$. This equation does not have a real solution!

Homework 5, Q2

In fact, $P < MC$ for any q . So the firm should shut down both in the short run and in the long run.

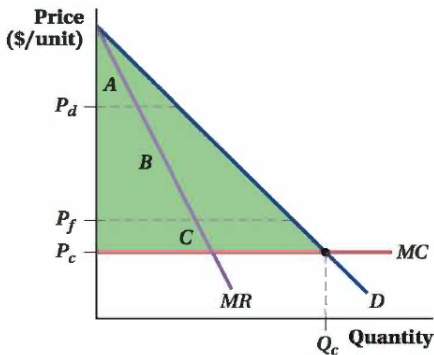


Perfect (“first-degree”) price discrimination

If the firm has complete information about every consumer's willingness to pay, the firm can charge each consumer the exact amount of their willingness to pay

- ▶ Producer captures all of the surplus; consumer surplus is zero

Consumer surplus	$= 0$
Producer surplus	$= A + B + C$
Deadweight loss from market power	$= 0$



Segmenting (“third degree” price discrimination)

If the firm knows there are groups of consumers with different demand curves, and can identify which customers have which demand curves, then the firm can offer different pricing to different groups

- ▶ e.g. student discounts
- ▶ e.g. senior citizen discounts
- ▶ e.g. lower drug prices in emerging markets

The differing characteristic(s) of each consumer group needs to be **observable** to the firm

- ▶ e.g. The firm can demand that consumers show student ID to enjoy student discounts

Third-degree price discrimination - exercise

A monopolist faces two group of consumers. Its first group of customers is represented by the demand equation $Q_1 = 10 - \frac{P_1}{2}$. The second group is represented by the demand equation $Q_2 = 10 - P_2$. Marginal cost is constant at \$8 in both markets. Assume zero fixed cost.

1. Suppose the firm can charge different prices to the two groups. Derive the profit-maximizing prices, quantities, and total profit.

Answer:

First group: $P_1 = 20 - 2Q_1$.

Total revenue $TR_1 = P_1 Q_1 = 20Q_1 - 2Q_1^2$. Marginal revenue $MR_1 = 20 - 4Q_1$. Profit maximization requires $MR_1 = MC$
 $\Rightarrow 20 - 4Q_1 = 8 \Rightarrow Q_1 = 3, P_1 = 14$. With constant MC, we have $MC = ATC$, so profit $\pi_1 = (P_1 - ATC)Q_1 = 18$.

Second group: $P_2 = 10 - Q_2$.

Total revenue $TR_2 = P_2 Q_2 = 10Q_2 - Q_2^2$. Marginal revenue $MR_2 = 10 - 2Q_2$. Profit maximization requires $MR_2 = MC$
 $\Rightarrow 10 - 2Q_2 = 8 \Rightarrow Q_2 = 1, P_2 = 9$. With constant MC, we have $MC = ATC$, so profit $\pi_2 = (P_2 - ATC)Q_2 = 1$.

Total profit $\pi = \pi_1 + \pi_2 = 19$.

Third-degree price discrimination - exercise

A monopolist faces two group of consumers. Its first group of customers is represented by the demand equation $Q_1 = 10 - \frac{P_1}{2}$. The second group is represented by the demand equation $Q_2 = 10 - P_2$. Marginal cost is constant at \$8 in both markets.

2. Suppose the firm has to charge the same price to both groups. Derive the profit-maximizing price, quantities, and total profit.

Answer: $P_1 = P_2 = P$, the firm chooses P to maximize profit under the condition that $Q_1, Q_2 \geq 0 \Rightarrow P \leq 10$.

Total market demand is $Q = Q_1 + Q_2 = 20 - \frac{3P}{2}$. Rearranging, we get $P = \frac{40-2Q}{3}$. Total revenue $TR = PQ = \frac{40Q-2Q^2}{3}$. Marginal revenue $MR = \frac{40-4Q}{3}$. Profit maximization requires $MR = MC \Rightarrow \frac{40-4Q}{3} = 8 \Rightarrow Q = 4, P = \frac{32}{3}$.

But with this price, group 2 buys a negative amount! This cannot happen. So, the firm will only sell to group 1. This becomes a simple monopoly problem with only group 1. From question (1), we have $P_1 = 14$. At this price, $Q_1 = 3, Q_2 = 0, \pi = \pi_1 = 18$.

Third-degree price discrimination - exercise

A monopolist faces two group of consumers. Its first group of customers is represented by the demand equation $Q_1 = 10 - \frac{P_1}{2}$. The second group is represented by the demand equation $Q_2 = 10 - P_2$. Marginal cost is constant at \$8 in both markets.

3. How does the total profit under single-price monopoly compare with that under segmenting? It is always true that segmenting cannot make the firm worse off than single-price monopoly?

Answer: Total profit under single-price monopoly (18) is smaller than that under segmenting (19).

It is always true, because single-price discrimination is just a special case of segmenting (where the firm happens to charge the same price for all consumer groups).

Menu pricing/Versioning (“second-degree” price discrimination)

If the firm knows there are groups of consumers, but cannot identify which customers belong to which groups, then the firm can offer a variety of price options for consumers to choose

This has to be **incentive compatible**: one group of consumer must not have an incentive to pose as another group to take advantage of the pricing options

- ▶ e.g. The firm cannot just offer two price options of \$5 or \$10 per unit — this way everyone will of course choose \$5

Ways of second-degree price discrimination:

- ▶ Menu pricing: e.g. quantity-based discounts
- ▶ Versioning: Offering different versions of the product (e.g. “Pro” vs “Standard”)

Homework 6, Q2

Pear Inc. sells high-end laptops to “Enthusiasts” and “Normals.” There is no way to charge different prices for the same product to the two groups.

It has developed two possible versions for this model year’s launch: Standard and Pro. Enthusiasts value a Pro version at 2200 and a Standard version at 1200. Normals value a Pro version at 1300 and a Standard version at 900. There are 10 millions total consumers. Let $x \in [0, 10M]$ be the number of Enthusiasts. The cost of production is 200 per laptop for either version.

Which versions should Pear offer, and at what prices, for each value of x ?

Homework 6, Q2

Enthusiasts value a Pro at 2200 and a Standard at 1200.

Normals value a Pro at 1300 and a Standard at 900.

There are $x \in [0, 10M]$ enthusiasts. MC is 200.

Answer:

There are three strategies tha the firm can choose:

1. Offer Pro only
2. Offer Standard only
3. Offer both Pro and Standard, with each type of product attracting different kinds of buyers

Homework 6, Q2

Enthusiasts value a Pro at 2200 and a Standard at 1200.

Normals value a Pro at 1300 and a Standard at 900.

There are $x \in [0, 10M]$ enthusiasts. MC is 200.

Answer:

Consider the first strategy, offering Pro only.

- ▶ At price $P_p \in (1300, 2200]$, only Enthusiasts will buy. The firm's profit is $(P_p - 200)x$, which is maximized at $P_p = 2200$. The firm's profit is then $(2200 - 200)x = 2000x$.
- ▶ At price $P_p \leq 1300$, both Enthusiasts and Normals will buy. The firm's profit is $(P_p - 200) \times 10M$, which is maximized at $P_p = 1300$. The firm's profit is then $(1300 - 200) \times 10M = 11000M$.

Consider the second strategy, offering Standard only.

- ▶ If sell to enthusiasts only, the best price is 1200, which is worse than offering only Pro to enthusiasts (at a price of 2200).
- ▶ If sell to everyone, the best price is 900, which is worse than offering only Pro to everyone (at a price of 1300).

Homework 6, Q2

Enthusiasts value a Pro at 2200 and a Standard at 1200.

Normals value a Pro at 1300 and a Standard at 900.

There are $x \in [0, 10M]$ enthusiasts. MC is 200.

Answer:

Third strategy: offering Pro and Standard to different kinds of consumers.

If we want Enthusiasts to buy Pro, and Normals to buy Standard:

- ▶ The firm's profit is $(P_p - 200)x + (P_s - 200)(10M - x)$
- ▶ We want the prices to be “incentive compatible”: Enthusiasts have no incentive to pose as Normals and buy Standard, and Normals have no incentive to buy Pro.
- ▶ This requires that Enthusiasts get higher consumer surplus from buying Pro than from buying Standard, and vice versa.
- ▶ For Enthusiasts: $2200 - P_p \geq 1200 - P_s \Rightarrow P_p - P_s \leq 1000$
For Normals: $1300 - P_p \leq 900 - P_s \Rightarrow P_p - P_s \geq 400$
- ▶ The highest prices to make this possible is $P_p = 1900$, $P_s = 900$.
The firm's profit is $1700x + 700(10M - x) = 7000M + 1000x$.

Homework 6, Q2

Enthusiasts value a Pro at 2200 and a Standard at 1200.

Normals value a Pro at 1300 and a Standard at 900.

There are $x \in [0, 10M]$ enthusiasts. MC is 200.

Answer:

Third strategy: offering Pro and Standard to different kinds of consumers.

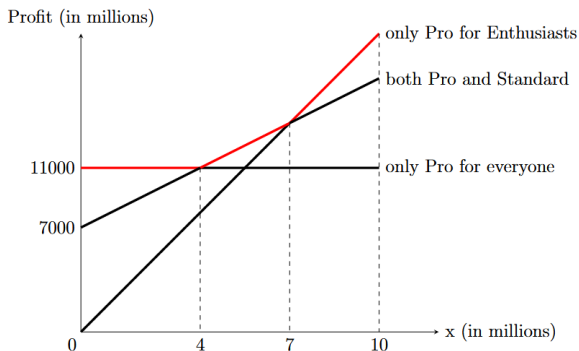
If we want Enthusiasts to buy Standard, and Normals to buy Pro:

- ▶ Incentive compatibility requires that Enthusiasts get higher consumer surplus from buying Standard than from buying Pro, and that Normals get higher consumer surplus from buying Pro than buying Standard
- ▶ For Enthusiasts: $2200 - P_p \leq 1200 - P_s \Rightarrow P_p - P_s \geq 1000$
For Normals: $1300 - P_p \geq 900 - P_s \Rightarrow P_p - P_s \leq 400$
- ▶ This is not possible!

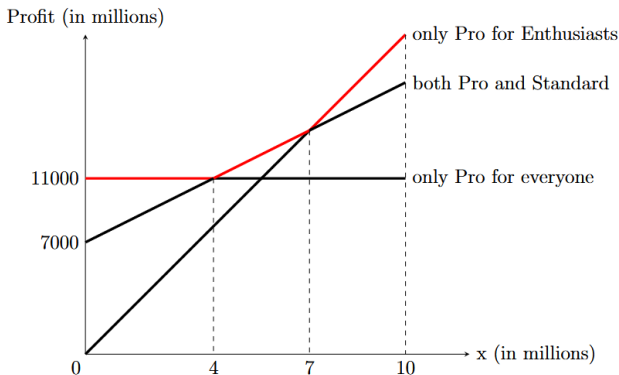
Homework 6, Q2

To sum up, here are the “valid” strategies and their profits:

1. Offering Pro only, at the price of \$2200, to Enthusiasts only:
Profit = $2000x$.
2. Offering Pro only, at the price of \$1300, to everyone:
Profit = $11000M$.
3. Offering Pro to Enthusiasts at \$1900, Standard to Normals at \$900:
Profit = $7000M + 1000x$



Homework 6, Q2



The best strategy is:

- ▶ If $0 \leq x \leq 4$: Offer Pro only, at the price of \$1300
- ▶ If $4 < x \leq 7$: Offer Pro at \$1900, Standard at \$900
- ▶ If $7 < x \leq 10$: Offer Pro only, at the price of \$2200